

ROLL THINNING VS SUPPORT WEAR

In bundles where vibration damage at the intermediate supports is a concern. Manufacturers will expand or roll the tubes at the supports to a metal-to-metal contact. This prevents movement.

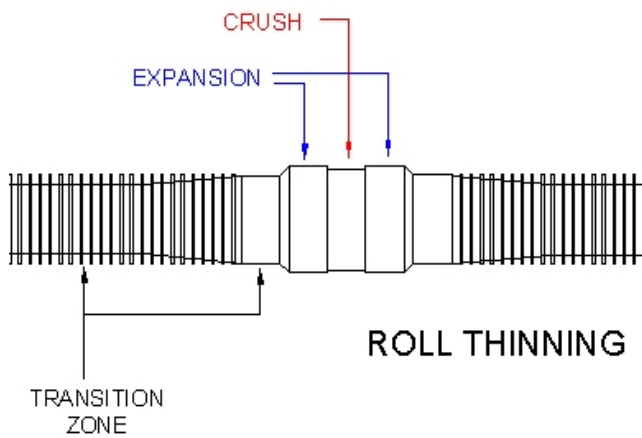


Figure 1

In the case of Roll Thinning, the tube is rolled and expanded beyond the metal to metal contact point. The tube is thinned 360° around the tube. The tube is locked in place and can not move, or vibrate. It may also be difficult to pull one out to replaced it. Rolling the tubes into the supports is a practice of the Trane corporation. They use it to prevent support wear in the Evaporator bundle of their chillers.

The problem with Roll Thinning is that it produces an indication SIMILAR to that of support wear. A well-trained technician will recognize the indication as Roll thinning by the expansion indication preceding the loss indication. See figure 2. Loss at a support of an un-expanded tube will be support wear. See Figure

3. Identification of the loss mechanism is critical so as not to condemn serviceable tubes. If you are testing a Trane unit you are probably looking at roll thinning. If you determine you are looking at roll thinning, use discretion in reporting it. Keep in mind that it is not an in-service wear and the tube has been in service from the beginning.

The rolling process crushes and displaces the tube material directly under the support. Unlike support wear, this is a nonprogressive thinning. It will not increase over time with machine use. Its common to see roll thinning of 3 divisions or more. Roll Thinning is typically about 4% - 7% per division on the cross axis channel. Assuming you are calibrate to QCP200. This means that roll thinning of 4 divisions is only a loss of between 16% and 28%. To our

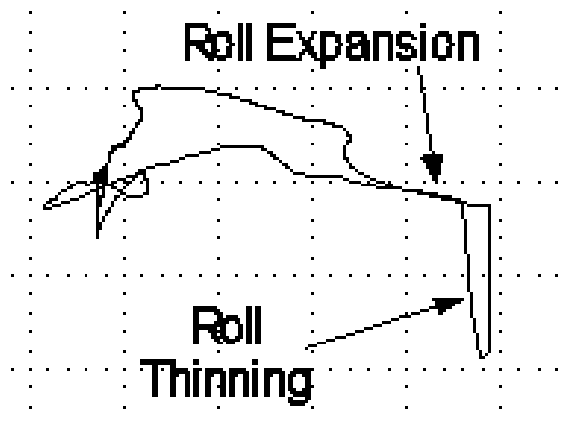


Figure 2

knowledge, there has never been a tube failure as a result of roll thinning. However, it is felt that excessive thinning, (8 Divisions or more) could initiate a circumferential crack.

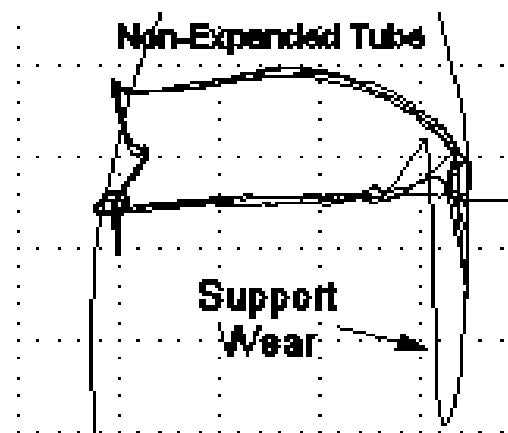


Figure 3

Support Wear is wear caused by the vibration of the tube in one or more of the intermediate supports. It is often initiated by corrosion which loosens the tube in the support hole allowing the tube to move and vibrate. The larger the hole becomes the more velocity the tube can gain before it strikes the surface of the support. That make this a very aggressive damage mechanism which will increase exponentially.

The loss is typically a 180° groove cut into the bottom side of the tube. This is due to the weight of the tube and its contents being lifted by the boiling action of the refrigerant as it changes state in the evaporator during normal operation.

Support wear will not occur in all tubes in a bundle. It is typically found in the upper rows of the evaporator. Indications in all tubes is a good indication you are looking at Roll Thinning.

Vibration can also be caused by the high velocity discharge gases in the condenser. This damage mechanism will result in the outer tubes closest to the gas having the greatest damage.

Support Wear on prime surface and skip fin tubes is best measured by its amplitude on the cross axis channel. Support wear on continuous fin tubes can also be measured by its amplitude. However, in cases where the support material is relatively less conductive than the tube material, phase analysis can be more accurate.

(Re: Bridgeport Tube Handbook, Pg 107) Over expanding or rolling the tube ends or land areas into oversized or out-of-round holes will stretch the circumference of the tube. This can result in high residual stresses immediately beyond the expanded section. The stressed region is usually 1/4" long and can develop into corrosion fatigue cracking, which accounts for most circumferential cracking.

A new tube placed in a tube sheet will be 3% - 5% smaller than the hole in the tube sheet. To hermetically seal the tubes in a chiller. The tubes are coated with a sealing compound such as Lock - Tite. The holes in the tube sheet are usually grooved to increase strength. The tubes are then rolled in the hole and crushed in the grooved hole another 15% - 20% beyond contact. This results in a 15% - 20% thinning of the tube wall at the bell end.

Roll thinning at the tube sheet. If you detect unusual indications of thinning at the tube sheet; determine if the loss is directly under the tube sheet or on the refrigerant side of the tube sheet. If it is thinning at the tube sheet, its due to the installation process, and will not change. Loss on the refrigerant side of the tube sheet, as shown in figure 4, is caused by in-service wear and should be reported. Report Roll Thinning only if it exceeds 3 divisions. If its detected in an isolated tube you should mark it as "Abnormal, Excessive." If several tubes are grouped together showing a loss pattern, you are likely looking at OD corrosion. You should mark them as OD Metal loss at the tube end as you would other OD metal loss.

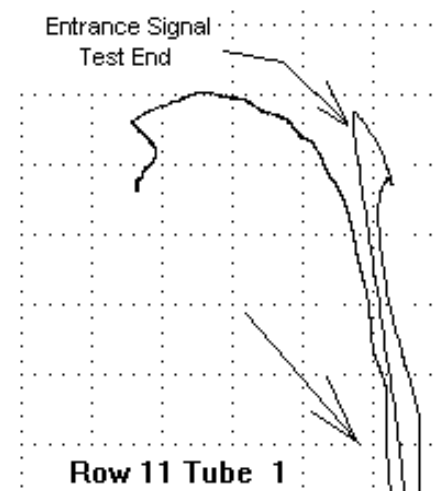


Figure 4